Geomatics and Geoinformation, MSc in Data Science A.Y. 2025-2026

EXERCISE 5: Geo Big Data handling and analysis - Machine Learning with Google Earth Engine

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June 2026

1 K-Means Clustering

Introduction

This section is about using the k-means algorithm to cluster different types of land and/or sea cover in an unsupervised way. The unsupervised framework is used in machine learning to make predictions about data without knowing their labels beforehand. In particular, clustering involves processing samples with the same set of features and grouping them based on the similarities between feature values. The k-means algorithm starts by assigning centroids randomly in space (the centers of the clusters), and mapping data points to the cluster associated with the closest centroid. Each centroid then gets updated, becoming the mean of the points that were mapped to its cluster. Iteratively, points are mapped to the new centroids, which get updated again, and the process continues until convergence or a stopping criterion is reached. In this exercise, data points are pixels of satellite images and their features are the reflectance values across different bands of the electromagnetic spectrum, as well as derived land surface temperature values.

Methods and Materials

Dataset. The dataset is the "USGS Landsat 7 Level 2, Collection 2, Tier 1". Landsat 7 is an Earth observation satellite launched by NASA and the United States Geological Survey (USGS) in 1999 as part of the Landsat program, a long-term mission dedicated to monitoring the Earth's surface. The satellite carries the Enhanced Thematic Mapper Plus (ETM+) sensor, which measures reflected and emitted electromagnetic radiation in visible, near-infrared, shortwave infrared and thermal infrared wavelengths. "Level 2" means that the dataset does not contain raw satellite measurements, but data that have already undergone atmospheric correction and other preprocessing steps to provide physically meaningful surface properties. It contains the surface reflectance of the visible bands SR_B1 (blue), SR_B2 (green), SR_B3 (red), near infrared SR_B4, shortwave infrared SR_B5 (SWIR 1) and SR_B7 (SWIR 2), and the surface temperature ST_B6. It also stores information about pixel quality in relation to cloud coverage and shadows.

Filtering. The region of interest selected for the analysis is Mount Fuji in Japan and some of its surrounding area, until the sea. A rectangular Polygon object was used to draw the area's geometry, with the following coordinates:

- 0: [138.4674723036692,35.03343656422262]
- 1: [139.3134195692942,35.03343656422262]
- 2: [139.3134195692942,35.54681958503143]
- 3: [138.4674723036692,35.54681958503143]
- 4: [138.4674723036692,35.03343656422262]

I chose this area because it contains different types of land cover, including vegetation, fields, urban locations, and water, offering a varied set of data points to train on. Images acquired between 1 January 2002 and 31 December 2002 were selected. Next, images were filtered by cloud cover, excluding scenes with more than 50% cloud coverage. The number of images returned after filtering was 44.

Radiometric Scaling. To save memory, reflectance and temperature values are not stored as floating point numbers but as integer digital numbers (DNs) in the dataset. For this reason, they must first be scaled in order to carry meaningful physical information. For the optical bands, the scaling follows the formula

$$\text{Reflectance} = \text{DN} \cdot 0.0000275 - 0.2.$$

For temperature values, the scaling is

$$\text{Temperature} = \text{DN} \cdot 0.00341802 + 149.0.$$

This conversion transforms the stored integer values into physically interpretable surface reflectance and land surface temperature measurements that can be compared across images and used reliably for clustering.

Cloud Masking. Every pixel contains QA (quality assessment) band values that are stored as bits. In particular, when cloud masking is performed, the focus is on bits 3 (cloud shadow) and 4 (cloud). Using Google Earth Engine's parallel computing map() function, each image is processed, and the QA band is inspected pixel by pixel. Pixels for which either the cloud or cloud-shadow bit is set are removed from the analysis, while only pixels whose cloud and cloud-shadow bits are equal to 0 are retained. This reduces the influence of atmospheric contamination on the clustering results.

Median-Aggregated Image. Next, a median image is generated from the pixel values of all images. The reason for choosing the median and not the mean is that the median is more robust to outliers. Otherwise, aggregated pixel values could be skewed if the data contained even a few pixels with extreme values. The result is an image with seven bands, the six surface reflectance bands and the surface temperature band. This aggregation is created so that the clusterer has a single, representative image to use as reference for pixel values and to reduce residual noise from clouds, shadows, or anomalous observations.

Creating Training Data and Training the Clusterer. In this step, 5000 pixels are sampled from the region of interest with a spatial resolution of 30 m, corresponding to the native spatial resolution of the Landsat 7 optical bands. It would be computationally inefficient to train a clusterer on every pixel in the image, so the sampled pixels provide a representative approximation of the region's spectral characteristics. Each sampled pixel is described by seven features corresponding to the six reflectance bands and the surface temperature band. A clusterer is then initialized with a chosen number of clusters k. I tested three different values of k: 3, 5, and 7. The clusterer is trained on the pre-processed data using k-means and subsequently used to classify all pixels in the median-aggregated image.

Results and Discussion

The images show the clustering results obtained using 3, 5 and 7 clusters, respectively. As expected, increasing the number of clusters increases the level of detail in the classification, allowing the algorithm to separate a greater number of spectrally distinct surface types. In all three classifications, water bodies are consistently assigned to a distinct cluster. The coastal waters surrounding the study area and the inland lakes are clearly separated from land surfaces. This behaviour can be explained by the spectral properties of water. Water strongly absorbs near infrared (NIR) and shortwave infrared (SWIR) radiation, which results in very low reflectance values in these bands compared to vegetation, soil and urban materials.

For the case of three clusters, the classification is relatively simple. One cluster corresponds to water, while the remaining two clusters broadly separate the terrestrial landscape into mountainous and vegetated regions on the one hand, and lower-elevation surfaces on the other. The latter include a mixture of urban areas, agricultural land and sparsely vegetated terrain. Although the result provides a clear overview of the main land-cover categories, the small number of clusters forces several different surface types to be grouped together.

With five clusters, the classification becomes more informative. Water remains clearly separated from land, while the forested slopes surrounding Mount Fuji are largely assigned to different classes than the lowland areas. Additional clusters appear within the terrestrial landscape, suggesting that the algorithm is distinguishing between vegetation, agricultural land, rocky mountain terrain and more densely developed areas. Although the classes do not perfectly correspond to physical land-cover categories, the resulting segmentation captures the major differences present within the study area while remaining relatively easy to interpret.

The seven-cluster classification gives the most detailed segmentation. We can see several additional classes appear across the mountainous regions and lowlands, indicating that the algorithm is separating subtle spectral

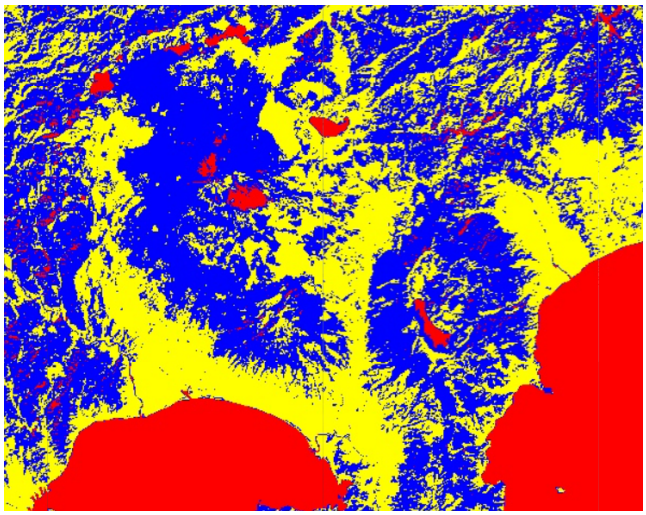
differences within previously unified classes. While this produces a more detailed representation of the landscape, it also introduces fragmentation, with visually similar surfaces often being assigned to different clusters. As a result, the interpretation of individual classes becomes less straightforward.

The snow-covered peak of Mount Fuji appears distinct in the original image and is assigned to different classes depending on the selected number of clusters. This behaviour is expected because snow has different spectral properties from vegetation, bare soil and urban materials.

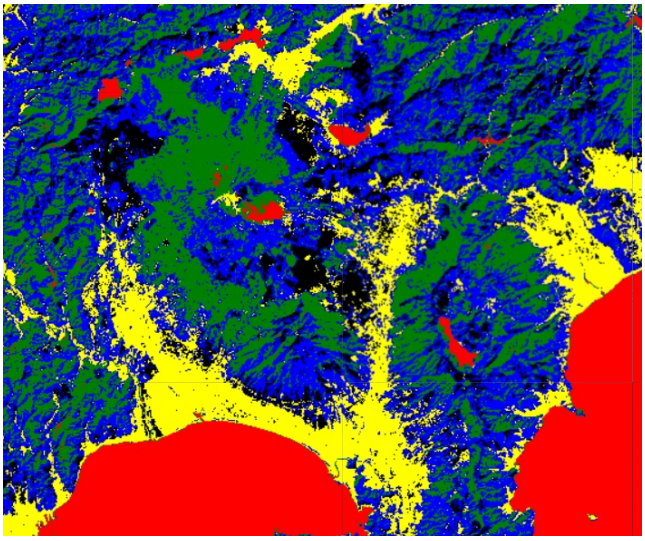
Overall, the results demonstrate that k-means clustering is able to identify most land cover types using only the spectral information contained in the Landsat imagery. Among the tested values, five clusters provide the best balance between simplicity and detail. The three-cluster solution is easy to interpret but merges various land-cover types into broad categories, while the seven-cluster solution introduces additional complexity without a corresponding increase in interpretability.



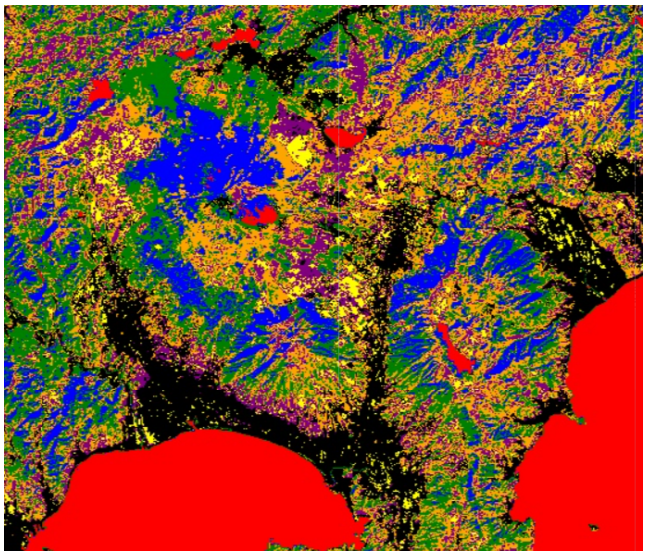
(a) Median-aggregated image, Mount Fuji (2002).



(b) K-means clustering with 3 clusters.



(c) K-means clustering with 5 clusters.



(d) K-means clustering with 7 clusters.

Figure 1: Comparison of the median-aggregated Landsat 7 image and the corresponding k-means clustering results obtained with different numbers of clusters.

2 Support Vector Machine

In this part of the project, the purpose is to use a support vector machine (SVM) to classify water and non-water pixels in a median-aggregated satellite image. A support vector machine is a machine learning model used in supervised learning for binary classification. It generally uses a hyperplane to separate instances of two classes into distinct groups, although variants of the SVM use curved surfaces to allow for non-linear splits.

Dataset. The dataset used for this task is "Landsat 8 Collection 2 Level-2, Tier 1 Surface Reflectance (SR)". It contains atmospherically corrected surface reflectance bands (five visible and NIR infrared bands and two short-wave infrared bands) as well as land surface temperature. The images also include intermediate and QA bands.

Data Preparation. The data was filtered based on date, region and cloud coverage. The region of interest chosen was Lake Victoria, in Africa, in order to have data on both water-covered areas and vegetation. These are the points of the geometry:

- 1. [33.95893665888134, -0.2804215586045142]
- 2. [33.95893665888134, -0.5543855850995356]
- 3. [34.40594288446728, -0.5543855850995356]
- 4. [34.40594288446728, -0.2804215586045142]

Only images from 2019 with cloud coverage of less than 50% were kept. After the filtering, 32 images were left. Radiometric calibration was carried out in the same way as in the previous point, using the scaling and shift parameters to turn the digital numbers into physically valid surface reflectance and surface temperature values. The cloud masking focused on the pixel QA bits 3 (cloud shadow) and 4 (cloud) to mask the clouds, in order not to influence later calculations. Next, the median-aggregated image was computed to create a representative image over the ROI.

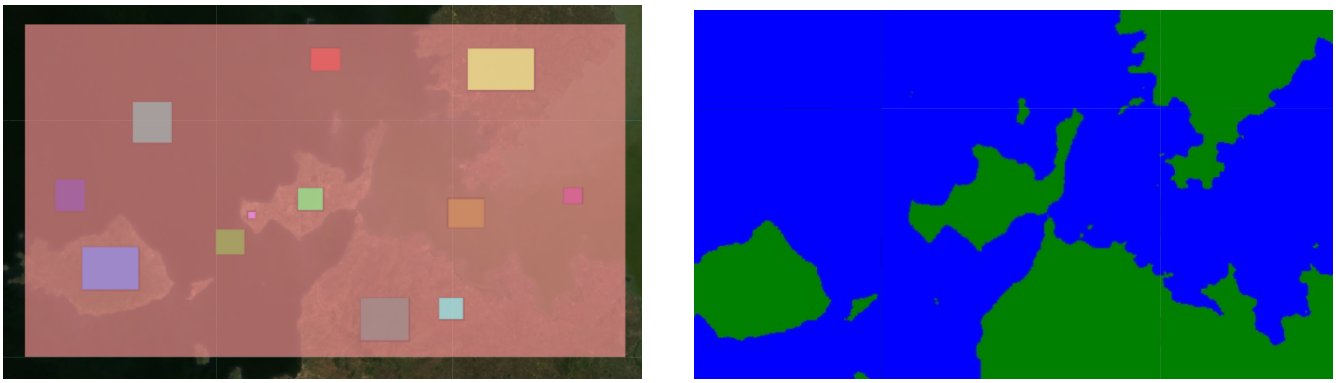
Training Data Preparation. Training data was manually generated by selecting polygons corresponding to two land-cover classes: water and non-water. Six rectangles were drawn over water surfaces and six rectangles over surrounding land areas. They were distributed across different portions of the lake and surrounding land areas in order to capture spectral variability within each class. These polygons were combined into a single FeatureCollection and assigned integer class labels, with water represented by class 0 and non-water represented by class 1. Information about the bands was extracted from all pixels contained within the training polygons using the sampleRegions() function in Google Earth Engine. Then, sampling was performed at the spatial resolution of 30m. Each sampled pixel was associated with its corresponding class label, generating a training dataset consisting of 102478 labelled observations. The features were selected based on the properties of water and its ability to absorb mostly in the NIR and SWIR regions of the EM spectrum, unlike vegetation and other land surfaces. The following variables were chosen as features: green (SR_B3), near infrared (NIR, SR_B5), shortwave infrared 1 (SWIR1, SR_B6), and 2 (SWIR2, SR_B7).

Support Vector Machine Classification. A supervised Support Vector Machine (SVM) classifier was implemented using the libsvm algorithm in GEE. In order to allow non-linear separation between the classes, the classifier was configured with a parameter gamma of 0.5 and cost parameter 10. This second parameter makes sure that false predictions get appropriately penalised, but not so much to risk overfitting. The model was trained using the labelled training dataset and the selected spectral bands. The trained SVM classifier was applied to the median composite image to generate a binary land-cover map. Each pixel within the ROI was assigned to either the water class (0) or the non-water class (1) according to its spectral characteristics. The resulting classified image was visualised using blue for water and green for non-water surfaces.

Results and Discussion

As seen in figure 2, the visual analysis of the classification map indicates a clear separation between water bodies and surrounding land surfaces. The classifier successfully identified the main lake area as water and correctly mapped the islands and shoreline regions as non-water. The selected spectral bands proved to be a suitable choice for distinguishing between water and non-water surfaces. Water strongly absorbs electromagnetic radiation in the Near Infrared (NIR) and Shortwave Infrared (SWIR) portions of the spectrum, resulting in low reflectance values

in bands B5, B6 and B7. In the meantime, vegetation and exposed land generally have higher reflectance in these wavelengths. The inclusion of the Green band (B3) helps to discriminate classes, as water typically exhibits relatively higher reflectance in the visible green region compared with the NIR and SWIR bands. The combination of these four bands therefore provided a robust feature space for binary water classification. Moreover, the classifier correctly identified several islands that were not explicitly represented by separate training classes. This shows the ability of the supervised learning approach to recognise land-cover types based on their spectral signatures rather than their geographic location. The successful classification of these features indicates that the model learned the underlying distinction between water and terrestrial surfaces.



(a) Training polygons used for SVM classification.

(b) Final SVM classification result.

Figure 2: Support Vector Machine (SVM) classification workflow for the Lake Victoria study area. (a) Training polygons used to generate labelled samples for the water and non-water classes. (b) Final binary classification map, where water is represented in blue and non-water surfaces in green.

3 Random Forest

In this last section, the supervised machine learning algorithm random forest is used to classify pixels into different land cover types. Random forest is an ensemble method that consists of multiple decision trees, which classify instances based on binary decisions operating on feature values. During training, the trees are constructed using random subsets of the training data and predictor variables. Each tree independently predicts a class for a given observation, and the final classification is determined by a majority vote among all trees. This ensemble approach aims to reduce overfitting and generally provides higher accuracy and robustness than a single land-cover mapping because it can handle large datasets, is relatively insensitive to noise, and performs well with multispectral satellite imagery.

Dataset and Filtering. The dataset for this task is the "Sentinel-2 Surface Reflectance (SR)" Harmonized image collection. Since Sentinel-2 is an optical satellite, this data does not include surface temperature measurements. The images were filtered by date, keeping only images from 2019 within the Camargue:

- 1: [4.363755612418161, 43.60494545625886]
- 2: [4.363755612418161, 43.33885028391176]
- 3: [4.752396969840036, 43.33885028391176]
- 4: [4.752396969840036, 43.60494545625886]

The Camargue represents a challenging classification environment due to its heterogeneous landscape, spanning wetlands, lagoons, agricultural fields, coastal beaches, salt marshes, and natural vegetation. For this reason, it is also an interesting ROI candidate to test the random forest on. To complete the filtering, images with a high cloud coverage (above 50%) were excluded. At the end, 195 images were left.

Data Preprocessing. Pixels containing clouds were masked, so that they would not affect the classification. The Sentinel-2 QA60 quality assessment band was used for this. In particular, bits 10 (cloud) and 11 (cirrus cloud) were

examined to create the mask. An alternative cloud masking approach based on the Cloud Score+ dataset was also tested to improve cloud detection performance. Since Sentinel-2 is an optical satellite, radiometric scaling was only performed on the digital numbers associated with reflectance. A scaling factor of 0.0001 was applied to all Sentinel-2 bands to convert the stored integers into physically meaningful surface reflectance values, approximately in the range [0,1]. Like in the previous two sections, the images were aggregated into a single representative image using the median reducer. This is useful for reducing the influence of residual clouds, shadows or other anomalies. To see if the image processing was successful, the aggregated image was visualised using both a true color (B4–B3–B2) and a false color (B11–B8–B3) composite. The first one is useful for visual interpretation, while the second can help understand the features that contribute to distinguishing between different land cover types.

Training Data. Random forest classification, like SVM, is a supervised machine learning problem, so when training a model, we need a dataset of samples with labels. For this exercise, the ESA WorldCover 2021 land cover product was used as the source of reference labels for classifier training and validation. The dataset contains eleven land cover classes at a spatial resolution of 10 m. In order to be processed by the classifier, the class values of the original dataset were remapped to a consecutive sequence of integer values ranging from 0 to 10. The remapped land cover layer was added as an additional band to the Sentinel-2 composite. Stratified sampling was then performed to extract 250 pixels from each land cover class within the ROI, for a total of 2000 samples. From this number, we can see that the ROI contains eight out of the eleven classes in total. This equal sampling from every class has the aim to remove bias and ensure that classes are balanced, regardless of the land cover of the specific ROI. A random value was assigned to each sampled pixel and used to divide the dataset into two independent subsets: 80% of the samples (1580) were assigned to the training dataset and 20% to the validation dataset (420).

Random Forest. A Random Forest classifier consisting of 50 decision trees was defined. It uses the spectral reflectance values of the Sentinel-2 bands as features to make predictions, and the ESA WorldCover classes represent the target labels. The model was trained to learn the relationships between spectral signatures and land cover classes. To measure the quality of the results, a confusion matrix and accuracy were computed on the training dataset and validation dataset. The first is to see how well the random forest was able to fit the training samples, while the second serves to measure the generalization abilities.

Land Cover Classification. The trained Random Forest classifier was then applied to the median-aggregated Sentinel-2 image. Each pixel was assigned to one of the land cover classes based on its spectral characteristics. The final classified image was visualised using a predefined land cover colour palette with hexadecimal colour codes, producing a thematic land cover map for the study area.

Results and Discussion

Visual inspection (figure 4) of the classified image shows that permanent water bodies (class 7, blue) were accurately detected within the Mediterranean Sea, coastal lagoons, and inland water bodies. Wetlands (class 8, cyan) were mainly distributed around the lagoon margins and transitional areas between land and water, which is consistent with the known ecological characteristics of the Camargue delta. Cropland (class 3, pink) occupies a large proportion of the eastern and northern portions of the study area, corresponding to the agricultural fields visible in the Sentinel-2 imagery. Built-up areas (class 4, red) were mainly identified around settlements and infrastructure networks. Bare or sparsely vegetated surfaces (class 5, grey) were concentrated along beaches, sand bars, and exposed coastal environments.

The confusion matrix obtained from the training dataset gave an overall accuracy of 99.30%, which is very high. The training confusion matrix shows that most classes were classified correctly, with very few misclassified samples. For example, tree cover, cropland, built-up areas, water bodies, and wetlands had nearly perfect classification rates:

$$\mathbf{CM}_{train} = \begin{bmatrix} 197 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 2 & 193 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 200 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 196 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 199 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 195 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 198 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 191 \end{bmatrix} \tag{1}$$

Such a high training accuracy indicates that the random forest model successfully learned the spectral characteristics of the training samples. However, training accuracy alone is not enough to evaluate model performance because the classifier is tested on data that were already used during model fitting. The independent validation dataset led to an accuracy of 70.48%. This value is much lower than the training accuracy, which indicates that the classifier does not generalise perfectly to new observations. Inspection of the validation confusion matrix reveals that several classes exhibit considerable overlap:

$$\mathbf{CM}_{val} = \begin{bmatrix} 31 & 11 & 5 & 2 & 1 & 0 & 0 & 0 & 1 \\ 6 & 34 & 4 & 2 & 3 & 0 & 0 & 0 & 5 \\ 2 & 7 & 19 & 6 & 3 & 4 & 0 & 0 & 7 \\ 0 & 0 & 5 & 40 & 6 & 0 & 0 & 0 & 2 \\ 0 & 0 & 1 & 2 & 44 & 2 & 0 & 0 & 2 \\ 0 & 0 & 1 & 1 & 5 & 47 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 46 & 2 \\ 2 & 5 & 3 & 2 & 2 & 7 & 0 & 1 & 35 \end{bmatrix} \tag{2}$$

The most frequent misclassifications occur between tree cover and shrubland, shrubland and grassland, grassland and cropland, wetlands and other vegetated classes. These errors are expected because these land cover types often have similar spectral signatures, especially when only spectral reflectance information is used, as in Sentinel-2 images. The confusion between these classes can be influenced even more by seasonal vegetation dynamics, mixed pixels, and transitional habitats.

One of the classes that was achieved the best performance was that of water bodies, with most validation samples being correctly identified. This result is expected because the spectral behaviour of water is more distinctive with respect to land, as we have also seen in the SVM case (especially in the near infrared and shortwave infrared bands). Built-up areas also achieved relatively good classification performance due to their strong contrast with vegetation and water surfaces.

However, there is a large difference between training and validation accuracies: $99.30\% - 70.48\% = 28.82\%$. From this discrepancy we can assume that there has been some overfitting. This could be explained by various factors. For example, the potential presence of errors or mixed pixels in the ESA WorldCover dataset that was used as reference data. We also have to consider that the Camargue region contains complex transitional environments between water, wetlands, agricultural land, and natural vegetation. These transitions can lead to gradual spectral changes that are challenging to separate into discrete classes. It's possible for some land cover classes to be underrepresented within the study area, reducing the classifier's ability to learn distinctive class characteristics. We can see this from the red and orange patches, especially. Another possible reason for the overfitting is that the model was trained using only reflectance bands. Additional variables that could help are vegetation or water indices such as NDVI and NDWI.



(a) True-colour median composite.



(b) False-colour median composite.

Figure 3: Median-aggregated Landsat imagery of the Camargue study area. (a) True-colour composite generated using the visible red, green and blue bands. (b) False-colour composite generated using near-infrared, red and green bands to enhance vegetation and water features.

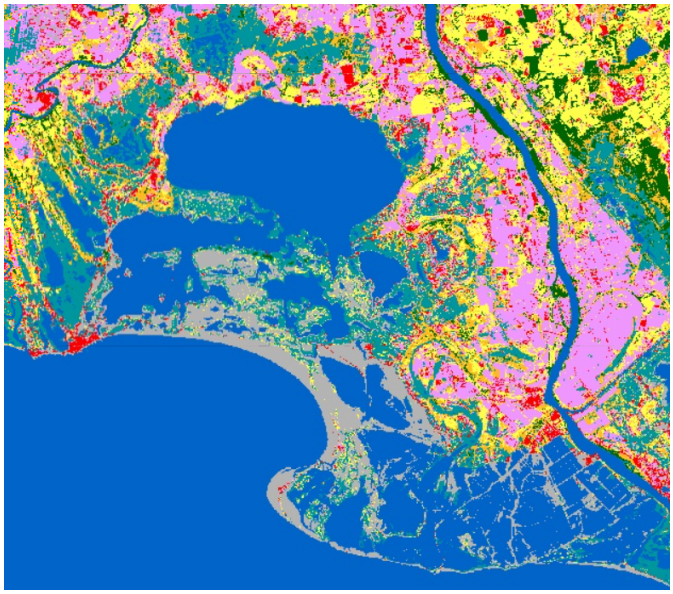


Figure 4: Random Forest land cover classification of the Camargue study area based on the 2023 Sentinel-2 median composite. Classes are: (0) Tree Cover, (1) Shrubland, (2) Grassland, (3) Cropland, (4) Built-up, (5) Bare/Sparse Vegetation, (6) Snow and Ice, (7) Permanent Water Bodies, (8) Herbaceous Wetland, (9) Mangroves, and (10) Moss and Lichen.